EARTH AS A PLANET

Topic 1: Introduction to Earth & The Solar System 🌞

- **Definition of Solar System:** The family of the sun, which includes the sun itself and the celestial bodies that revolve around it.
- **Key Points:**
    - Earth is our home planet. It is the **third planet** from the sun and the **fifth-largest** in the solar system.
    - It is the absolute *only* planet known to support life! 🌱
    - From space, Earth looks blue because 70% of its total area is covered by water. Because of this, it is called the **"watery planet"**. 🌊
    - The chief members of the solar system are planets, satellites (moons), asteroids, comets, and meteorites.
    - The planets in order of distance from the sun are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune.

What does it basically mean? 🧐

Think of the Solar System as a big cosmic family where the Sun is the parent and all the planets are the kids spinning around it! Earth is the lucky third child because it sits in the perfect spot to have oceans of water. Because there is so much water reflecting light, astronauts looking at Earth from space just see a beautiful, giant blue marble! 🌎✨

Topic 2: The True Shape of the Earth 🌎

- **Definition of Oblate Spheroid (Geoid):** The true shape of the Earth, which is not a perfect sphere but bulges slightly at the center (Equator) and is flattened at the top and bottom (Poles). "Geoid" simply means "earth-shaped".
- **Key Points:**
    - In the olden days, people thought the earth was a flat disc with steep edges (they were afraid of falling off!).

- **Aryabhatta**, a 5th-century Indian astronomer, and **Pythagoras**, a 6th-century Greek philosopher, were the first to say the earth is round and rotates.
- The Earth's diameter at the Equator is **12,756 km**, while at the Poles it is **12,714 km**.
- This 42 km difference is caused by the **centrifugal force** of the earth spinning at great speed, which squishes the poles and pushes the middle out!

What does it basically mean? 🤔

Imagine holding a soft squishy ball and pressing down on the top and bottom with your hands. The middle part will puff out, right? That is exactly what happened to Earth! Because it spins so fast like a top, the middle belly (Equator) bulges outward, making it look slightly chubby instead of being a perfectly round basketball! 🏀➡️🍊

Topic 3: Proofs that the Earth is Spherical 🚢🔭

- **Definition of Proofs:** Real-world observations and scientific experiments that confirm the earth is round, not flat.
- **Key Points (The 9 Major Proofs):**
  - **1. Sighting of a ship:** When a ship comes toward the shore, we first see the top mast, then the deck, and finally the bottom hull. If the earth were flat, we would see the whole ship at once! 🛳️
  - **2. The Bedford Level Experiment:** Three equal poles were placed 5 km apart in a canal in England. Through a telescope, the middle pole looked slightly higher. This only happens on a curved surface!
  - **3. Pole Star:** The Pole Star is seen at a 90° angle at the North Pole, but the angle decreases as you go to the Equator (where it is 0°). This shifting angle proves the surface is curved. ⭐
  - **4. Satellite Pictures:** The most conclusive evidence! French astronomer Jean Baptiste Delambre discovered the flattened poles, and 20th-century satellite photos from space finally proved it. 🛰️

- **5. Lunar Eclipse:** During an eclipse, the shadow of the earth on the moon is *always* circular. Only a sphere casts a perfectly round shadow from every angle! 🌕 🌑
- **6. Circular Horizon:** When you climb a high mountain or fly in a plane, the horizon looks like a giant circle that gets wider the higher you go. ⛰️
- **7. Sunrise and Sunset:** Because the earth is round and spins West to East, people in the East (like Japan or India) see the sunrise earlier than people in the West (like America). If it were flat, the sun would rise everywhere at the exact same time! 🌅
- **8. Circumnavigation:** In 1522, Ferdinand Magellan's ship named *Victoria* sailed all the way around the world and returned to Spain where it started without falling off an edge! ⛵
- **9. Earth as a Heavenly Body:** Every other planet in the solar system is round because of continuous rotation in space. Earth belongs to this same family, so it cannot be a flat exception! 🪐

What does it basically mean? 🧐

For a long time, people argued about whether the Earth was a flat table or a round ball. Slowly, smart observers realized things didn't make sense for a flat world. Why do ships seem to sink bottom-first over the horizon? Why do shadows on the moon look round? Finally, when explorers sailed around it and astronauts took photos from space, the debate was closed forever: Earth is a big, beautiful spinning sphere! 📸 🌎

Topic 4: Earth as the Home of Humankind (Distance & Temp) 🌡️

- **Definition of Habitable:** Having the right conditions (like temperature and distance from the sun) to support living things.
- **Key Points:**
  - **Distance from the Sun:** Earth is at an *optimum* (perfect) distance. If we were closer, everything would burn up. If we were farther away, everything would freeze into solid ice! 🧊 🔥

    - **Temperature:** Earth has a lovely average temperature of **17°C**, which is just right for life.
    - If the average temperature changes by just a few degrees, many plant and animal species would perish (die off).
    - Other planets are extreme: Mercury and Venus are boiling hot (over 400°C), while planets past Mars are freezing cold (below 0°C).
- **Extra Fact:** Venus is actually hotter than Mercury (even though Mercury is closer to the sun!). This is because Venus has a thick atmosphere made of carbon dioxide, which traps heat like a giant oven (Greenhouse Effect)! 🌫️

What does it basically mean? 🤔

Earth is basically the "Goldilocks" of the Solar System! It's not too hot like Venus, and it's not too cold like Mars—it is *just right*. Because we sit at the perfect distance from the Sun's campfire, our weather is comfortable enough for humans, plants, and animals to thrive without burning up or freezing solid! 🐻🥣

Topic 5: The Atmosphere and Water 💧☁️

- **Definition of Atmosphere:** The protective layer of air surrounding the earth, made up of life-supporting gases.
- **Key Points:**
    - **Composition of Air:** Nitrogen (78%), Oxygen (21%), Carbon Dioxide, Helium, and Argon (less than 1% combined).
    - **Temperature Balance:** The atmosphere receives heat from the sun and loses heat by radiating it back into space, keeping a perfect balance so we stay warm. 🌞
    - **Ozone Layer:** A special gas in the atmosphere that acts like sunscreen, absorbing harmful ultraviolet (UV) rays from the sun. 🧴
    - **Water:** 70% of Earth is water. Water distributes heat and moderates the climate so the surface condition is stable.

What does it basically mean? 🧐

Imagine the atmosphere is a giant, cozy blanket wrapped around the Earth. It gives us oxygen to breathe and traps just enough of the sun's heat so we don't freeze at night! Meanwhile, the oceans act like giant air conditioners, moving warm and cold water around the globe to make sure the weather stays friendly for us to live in! 🌬️🌊

Topic 6: Biosphere, Ecosystem, and Lithosphere 🌿🐾

- **Definition of Ecosystem:** A self-regulating and self-sustaining structural and functional unit of the biosphere (like a pond, desert, or forest) that depends on the sun for energy.
- **Definition of Biosphere:** The vast network of all interconnected ecosystems. It is the "life zone" of the earth where air, land, and water meet so life can develop.
- **Definition of Hydrological Cycle (Water Cycle):** The continuous movement of water from the oceans to the atmosphere (evaporation), and back down to the land as rain (precipitation).
- **Definition of Lithosphere (Solid Crust):** The solid outer crust of the earth.
- **Key Points:**
    - Everything in nature is interconnected. Materials and energy are constantly exchanged within an ecosystem. ♻️
    - **Water's Superpower:** Water can absorb an enormous amount of heat during the day without getting too hot itself. At night, it releases that heat to keep the earth from freezing. Without water, our days would be boiling and our nights would be freezing! 🧊🔥
    - **The Crust:** Weathering (breaking down) of rocks on the Lithosphere creates soil. Soil provides the nutrients necessary for plants to grow, which in turn feeds all the animals! 🦎🌻

What does it basically mean? 🧐

The Earth is the ultimate team project! The **Lithosphere** (the hard ground) provides the soil and nutrients. The **Hydrosphere** (water) rains down to give everything a drink. The **Atmosphere** (air) gives breath and warmth. When these three team up, they create the **Biosphere**—the magical zone where bugs, birds, trees, and humans get to live and play together in one giant, beautiful ecosystem! 🤝🌍

Here are the final, super easy-to-understand notes for the rest of Chapter 1! Let's finish strong with lots of clear points and fun emojis! 🌟🌍

Topic 7: The Biosphere (The Realm of Life) 🌱

- **Definition of Biosphere:** It is the narrow, special realm of contact and interaction between the atmosphere (air), lithosphere (land), and hydrosphere (water).
- **Key Points:**
    - The biosphere provides absolutely all the necessities for species to live on earth, including light, heat, water, food, and habitats. ☀️💧
    - It is a very thin layer of approximately **15 km** total, stretching from the deepest ocean trench all the way to the highest mountain peak! ⛰️
    - At ground level, it only extends to a depth of just **3 meters** below the soil.
    - In ocean waters, life is mostly found in the top **200 meters** where marine and freshwater creatures live. 🐟
    - Within the biosphere, life is chiefly found in two forms: the **plant kingdom** and the **animal kingdom**. 🦁🌳

What does it basically mean? 🧐

Imagine a giant, lifeless rock floating in space. Now, paint a super thin, magical layer of water, dirt, and air right on the very surface of that rock. That incredibly thin layer where the air, land, and water touch each other is the *only* place where life can exist! It's the ultimate "sweet spot" that gives us our food, our homes, and our perfect weather. 🏡✨

Topic 8: Life-Giving Cycles & The Nutrition Cycle ♻️🍽️

- **Definition of Nutrition Cycle:** A cycle representing the relationship and interdependence between living and non-living things in our environment.
- **Key Points:**
    - Various life-giving cycles operate in nature strictly to balance the factors necessary for life. ⚖️
    - The nutrition cycle operates through the food chain and processes like photosynthesis, transpiration, and respiration.
    - **Producers:** Plants are the producers of food. Energy starts here through a process called photosynthesis. 🌻
    - **Consumers:** These are the herbivores (plant-eaters) and carnivores (meat-eaters) who live off the producers. 🐇🐅
    - **The Top Consumer:** Man is at the very top of this chain because humans can modify their environment and create changes at any level of the food chain! 🧍
    - **Decomposers:** All living beings eventually undergo decomposition by the action of saprotrophs (decomposers like fungi and bacteria). 🍄

What does it basically mean? 🤔

This is the ultimate "circle of life"! Nothing in nature works alone. The sun gives energy to the plants to make food. The animals eat the plants, and bigger animals eat the smaller ones. When things die, tiny bugs and mushrooms break them down into dirt, which feeds the soil so brand new plants can grow again! It is a perfect, never-ending recycling machine. 🔁🐾

Topic 9: The Carbon Cycle ⬛🔥

- **Definition of Carbon Cycle:** The cyclic movement of carbon from the atmosphere to organisms, and from organisms back to the atmosphere.
- **Key Points:**
    - Carbon is the basic raw material of all life! Every single living thing contains carbon. 🧬

- **Step 1 (Taking it in):** Atmospheric carbon dioxide (CO_2) is used by plants (along with sunlight and chlorophyll) to make food. The CO_2 is fixed as starch. 🌿
- **Step 2 (Moving it around):** Carbon moves along the food chain as animals eat the plants.
- **Step 3 (Going underground):** After death, plants and animals are decomposed by saprophytic bacteria. If they get buried deep in the soil and undergo slow compaction over millions of years, they turn into **fossil fuels** containing huge amounts of carbon! 🛢️
- **Step 4 (Releasing it):** During respiration (breathing), plants, humans, and animals use oxygen and release CO_2 back into the air. Burning fossil fuels also releases massive amounts of CO_2. 🏭
- **The Big Warning:** Any imbalance of carbon in the atmosphere (like using too many fossil fuels) heats up the earth and leads to **Global Warming**. 🌡️🥵

What does it basically mean? 🧐

Carbon is like a magic building block that travels everywhere. It floats in the air, plants breathe it in to grow, we eat the plants to get strong, and when we breathe out, we blow it right back into the air! If a plant gets buried for a really long time, it turns into coal or oil. Nature has a perfect budget for this carbon, but when humans burn too much coal and oil, we put too much carbon back into the air, which traps heat and makes the whole planet run a fever! 🥵🌎

Topic 10: Nitrogen and Oxygen Cycles 🌬️

- **Definition of Nitrogen Cycle:** The cycle where nitrogen moves from the atmosphere to the soil, through the food chain, and back to the soil/air.
- **Definition of Oxygen Cycle:** The cycle where living things take in oxygen to release energy from food, and green plants release it back into the atmosphere.
- **Key Points (Nitrogen Cycle):**

    - The atmosphere is the largest reservoir of free nitrogen (approximately 78%). ☁️
    - Plants cannot breathe it in! Instead, they take nitrogen from the soil in the form of **nitrates**. 🌱
    - It moves through the food chain to animals, and is returned to the soil through manure, excreta, dead bodies, and micro-organisms.
- **Key Points (Oxygen Cycle):**
    - Living things use oxygen to release energy from the food they eat. 🏃
    - Oxygen is used together with carbon, hydrogen, and nitrogen to build new molecules in our bodies. 🧱
    - It is released back into the atmosphere by green plants during photosynthesis.

What does it basically mean? 🧐

Even though the sky is completely full of nitrogen gas, plants can't just breathe it in. They have to "drink" it from the dirt after special bacteria turn it into a yummy nutrient soup called nitrates! Meanwhile, oxygen is the fuel that keeps our bodies running. We breathe it in to burn our food for energy, and we breathe out carbon dioxide. The plants do the exact opposite, breathing in our waste and breathing out fresh oxygen for us. These cycles are nature's brilliant mechanism to maintain total balance! ⚖️🌳

Topic 11: Important Terms to Remember (Glossary) 📖

🧠 Term	📝 Definition
Lithosphere	The solid layer of rocks on the surface of the earth. 🪨
Satellite	A celestial body that revolves around a planet. 🌕

🧠 Term	📝 Definition
Circumnavigation	The act of navigating (sailing) completely around the globe. ⛵
Atmosphere	The protective blanket of air surrounding the earth. ☁️
Photosynthesis	The process by which green plants make their own food using sunlight, carbon dioxide, and water. 🌻
Greenhouse Effect	The trapping of the sun's radiation by gases (like carbon dioxide) present in the lower atmosphere, making the planet hotter. 🌡️

Question and Answers

I. Choose the correct option: 📝

1. At the Bedford level canal area a famous experiment was carried out. How many poles were used and at what distance apart? 📏

(a) 3 poles, 5km ✅

2. You are inside a hot air balloon floating at 500 m above the ground. What would you say about the view of the earth's surface as you see it? 🎈

(c) circular ✅

3. Assertion (A): With increase in altitude the circular horizon of the earth's surface also widens. Reason (R): The earth's surface is flat. 🌰

(c) A is true but R is false. ✅

4. At what angle can the Pole Star be seen at Macapa, Brazil? [Hint: It lies on the equator] ⭐

(d) 0° ✅

5. Assertion (A): The Pole Star lies in line with the axis of the earth with its angle decreasing towards the Equator. Reason (R): The earth is spherical in shape. 🌍

(a) Both A and R are true and R is the correct explanation of A. ✅

6. In 1522 Magellan's ship Victoria returned to Spain. What did this voyage prove? ⛵

(a) The earth is spherical ✅

7. Ancient cultures took earth for a disk, but very soon the observation of stars and planets revealed that earth resembles a sphere. After the late 1700s measurements showed that earth is slightly flattened at the poles. What is the actual shape of the earth? 🔭

(d) Oblate Spheroid ✅

8. Given below are pairs of statements. Select the statements which prove the spherical shape of the earth. (1. If there were no buildings... 2. The boat would stay in view... 3. A ship slowly starts to disappear... 4. If we were to place two identical sticks... 5. During a lunar eclipse...) 🚢

(c) 3, 4, 5 ✅

9. Which gas dominates the atmosphere of Venus? 🌫️

(b) Carbon dioxide ✅

10. Earth happens to orbit the sun within the so-called "Goldilocks zone," where temperatures are just right to maintain liquid water on our planet's surface. Water covers roughly 71% of Earth's surface, with most of that water located in our planet's _____. 💧

(c) Oceans ✅

11. Assertion (A): Water on earth drastically tones down day and night temperature variation. Reason (R): Water can absorb enormous amount of heat without causing much change in its temperature. 🌊

(a) Both A and R are true and R is the correct explanation of A. ✅

12. Name the gas that absorbs the harmful ultraviolet rays of the sun. ☀️

(a) Ozone ✅

13. Assertion (A): The earth is the only planet known to support life. Reason (R): The earth is a terrestrial planet like Mercury and Venus. 🌱

(b) Both A and R are true but R does not explain A. ✅

14. Assertion (A): Earth is a watery planet that supports life. Reason (R): Distribution of water is responsible for moderating the surface condition of the earth. 🌍

(a) Both A and R are true and R is the correct explanation of A. ✅

15. Earth has a very hospitable temperature and mix of chemicals that have made life abundant here. Most notably, earth is unique in that most of our planet is covered in liquid water, since the temperature allows liquid water to exist for extended periods of time. Besides liquid water and temperature identify the other features of our planet that make it great for sustaining life? 🏠

(d) All of the above. ✅

16. _____ is a self-regulating and self-sustaining unit of the biosphere? 🌿

(a) Ecosystem ✅

17. Name the cycle through which solar energy moves from non-living to living organisms and back. ♻️

(c) Nutrition cycle ✅

18. Your teacher is holding a piece of coal and says it contains the basic raw material for life. What does coal comprise? 🪨

(b) Carbon ✅

19. Name the genius Indian mathematician astronomer from the ancient world who was the first proponent of the round shape of the earth. 📜

(b) Aryabhatta ✅

20. You are standing at a beach looking towards the horizon. You hear the horn of a ship. Which part of the ship will you see first? 🚢

(c) Mast ✅

21. What is the thickness of the Biosphere? 🏔️

(c) 15 km ✅

II. Short Answer Questions ✍️

1. How can you prove that the earth is a sphere by looking at the horizon? 🌅

Looking at the horizon reveals a circular shape, and as you go higher up in altitude, this circular horizon widens even more, which proves the earth's surface is curved like a sphere! 🌍✈️

2. Briefly describe the shape of the earth. What is the specific name given to it? 🍊

The earth bulges out slightly at the center (the Equator) and is flattened at the top and bottom (the poles). This unique, slightly squished shape is specifically called an Oblate Spheroid (or Geoid)! ✨🔴

3. What is the earth's mean temperature? State its one advantage. 🌡️

The earth's mean temperature is a very comfortable 17°C! The biggest advantage of this temperature is that it is hospitable enough to allow liquid water to exist for long periods, which is absolutely essential for sustaining all life! 💧🌻

4. Why is the earth called a watery planet? 🌊

The earth is known as a watery planet because a massive 70% of its total surface area is completely covered by water, primarily in its huge blue oceans! 🐋🌎

5. What is 'biosphere'? 🌱

The biosphere is the super thin, self-regulating "life zone" of the earth where the atmosphere (air), lithosphere (land), and hydrosphere (water) all touch and interact to provide everything living species need to survive! 🐾🌳

6. Name the conditions necessary for life on earth. 🏡

Life on earth thrives perfectly because of a few key conditions: a hospitable temperature, the protective atmosphere (with oxygen and ozone), the optimal

distance from the sun, the solid lithosphere for soil, and the abundant presence of liquid water! 🌞💧

7. Give any two features of the earth that make it a 'Unique Planet'. ⭐

Two amazing features that make Earth unique are its abundance of liquid water (the Hydrosphere) and its perfect "Goldilocks" temperature that supports a rich Biosphere full of plants and animals—something no other known planet has! 🌸🦓

8. What would happen if the average temperature of the earth increased by half a degree? 🥵

If the average temperature increased by even a tiny amount, it would disrupt the delicate climate balance, causing extreme heat or cold variations that would lead to many plant and animal species dying off! 🥀🔥

9. Trace the cyclical movement of carbon in the three realms of the earth. 🔄

Carbon is taken right from the atmosphere by plants in the biosphere to create food. When plants and animals eventually die, they decompose deep into the soil of the lithosphere, sometimes forming fossil fuels. Finally, it gets released back up into the atmosphere when organisms breathe or when those fossil fuels are burned! ⬛🌫️

1. (a) How does the sighting of a ship from the seashore prove that the earth is round in shape? 🚢

When you watch a ship coming towards the shore, you don't see the whole ship all at once! First, you see the tall mast at the very top, then the middle deck, and finally the bottom hull. 🌊 This step-by-step appearance only happens because the earth's surface is curved. If the earth were totally flat, the entire ship would just appear small and get bigger all at once as it came closer! 🌍✨

1. (b) Use a 1m long table to replicate the Bedford Level Experiment. Briefly mention how you carry out the experiment to prove the shape of the earth. 📏

To replicate this, you would actually need a slightly curved table (since a completely flat table won't show the earth's curve). For the real experiment, you place three poles of the exact same height in a straight line at equal

distances. 🔭 When you look through a telescope from one end, the middle pole looks slightly higher than the other two! This proves the surface they are resting on is a gentle curve, not a perfectly flat line! 🌰👀

1. (c) Give a geographical reason for each of the following:

- **(i) The sunrise and sunset occur at different times at different places.** 🌅

Because the earth is a spinning sphere! As it rotates from West to East, the places in the East face the sun first and get an early sunrise. The places in the West are still hiding in the dark and have to wait their turn! 🌍🌞

- **(ii) The earth is not a perfect sphere.** 🔴

The earth spins on its axis at a super fast speed! This spinning creates a strong force that squishes the earth at the top and bottom (the poles) and pushes its belly out in the middle (the Equator). That's why its true shape is an "Oblate Spheroid"! 🌪️🌍

- **(iii) Venus is hotter than Mercury.** 🔥

Even though Mercury is closer to the sun, Venus is hotter because it has a super thick atmosphere completely full of carbon dioxide. This gas acts like a giant, heavy blanket that traps the sun's heat inside (the Greenhouse Effect) and won't let it escape! 🌫️🌡️

1. (d) Draw a well labelled diagram to show that the horizon of the earth is circular in shape. 🖍️

(Since I cannot physically draw, here is how you can easily sketch it!) ✍️ Draw a large curved line representing the Earth. On top of the curve, draw a little stick figure standing up. Draw straight dashed lines going out from the person's eyes so they barely touch the edges of the curve. Label the person as "Observer" and the points where the dashed lines touch the earth as the "Circular Horizon." This perfectly shows how our view is blocked by the curve! 🏔️👁️

2. Describe the role of each of the following in making earth a habitable planet. 🏡

- **(a) Atmosphere:** ☁️ It acts like a magical protective blanket around the earth! It gives us oxygen to breathe, traps just enough heat to keep us

warm, and has a special ozone layer that blocks harmful ultraviolet rays from the sun. 🛡️🌬️

- **(b) Water:** 💧 Water covers 70% of the earth and is essential for all living things to survive and grow! It also absorbs the sun's extreme heat during the day and releases it at night so our planet's climate stays friendly. 🌊🌡️
- **(c) Temperature:** 🌡️ Earth has a perfect average temperature of 17°C. It is the only temperature where water can exist as a flowing liquid (not just boiling gas or solid ice), which is the most important requirement for life to exist! 🧊🔥
- **(d) Distance from the Sun:** 🌞 Earth sits in the perfect "Goldilocks zone." We aren't too close to the sun (so we don't burn up) and we aren't too far away (so we don't freeze). The distance is just absolutely right for us! 🐻🥣

3. (a) How does each of the following prove the circular shape of the earth:

- **(i) Lunar Eclipse:** 🌕🌑 During a lunar eclipse, the earth passes right between the sun and the moon. The dark shadow that the earth casts on the moon is always perfectly circular! Only a round, spherical object can cast a round shadow from every single angle. 🌑🔭
- **(ii) Pole Star:** ⭐ The Pole Star is seen directly overhead at a 90° angle if you stand at the North Pole. But as you travel south toward the Equator, the angle slowly decreases all the way to 0°. This smooth, changing angle only happens if you are walking down a curved surface! 🚶‍♂️📐

3. (b) What is an ecosystem? Give an example. 🌿

An ecosystem is a self-sustaining community where living things (like plants and animals) interact with non-living things (like soil, water, and sunlight) to survive together in perfect balance. A great example is a **Forest Ecosystem**! 🌳🦌

3. (c) Give a geographical reason for each of the following:

- **(i) Earth is a habitable planet.** 🌱

Earth has the absolute perfect combination of a solid crust covered with green vegetation, an atmosphere full of breathable air, and plenty of liquid water. This amazing mix provides all the necessities for life! 🌍💚

- **(ii) From space Earth looks blue.** 🛰️

When astronauts look at Earth from way up in space, they see a giant blue marble because a massive 70% of our planet's surface is covered by enormous blue-green oceans! 🌊🌎

- **(iii) Water bodies reduce the day-night temperature variations.** ⚖️

Water has an amazing superpower: it can absorb a huge amount of heat during the sunny day without getting too hot itself. At night, it slowly releases that trapped heat back into the air. This keeps the coastal areas from becoming freezing cold at night or boiling hot in the day! 🌞🧊

3. (d) Draw a well labelled diagram to show the cyclic circulation of water between the three realms of the earth. 🖍️

(Here is how to draw this fun cycle!) ✍️ Draw a picture with a blue ocean (Hydrosphere), land with green trees (Lithosphere), and fluffy clouds in the sky (Atmosphere). Draw squiggly arrows pointing UP from the ocean and trees to the clouds and label it "Evaporation & Transpiration." Draw an arrow pointing DOWN from the clouds to the land and label it "Precipitation (Rain)." Finally, draw an arrow on the land pointing back into the ocean and label it "Surface Runoff." 🌧️🌲🌊